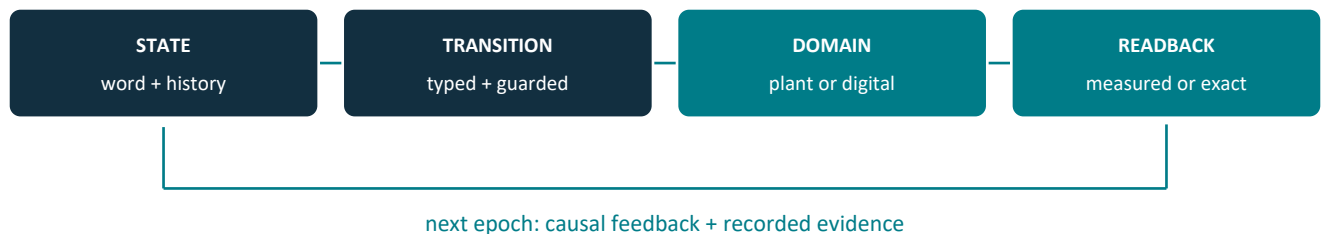


PERPETUUM MOBILE

A formalization of formalizations for analog, digital, physical and future application architectures



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Source-framework attribution: NL200678942

One formal language. Seven substantive applications.

Seeded worlds; analog material computation; adaptive robotic surfaces; programmable wave media; feedback fabrication; regenerative habitats; distributed embodied systems.

PM 1.0 / 15 September 2026

A new synthesis of WANTWOMBAN, atomOS v3.6 M1, its K1 device report, and UGTS-KC 3.6. Existing release labels remain unchanged.

Prepared with Codex computational assistance. Source attribution is retained; the new application architectures, adapters and proofs are explicitly identified as additions. The four supplied PDFs and editable specification are embedded as attachments.

01 / INTENT

What this document defines

Perpetuum Mobile is a family of systems that continually observe, transform and reconstitute state. The common object is a typed, causal loop. Its embodiments range from reproducible digital spaces to dissipative material computers and resource-limited physical installations.

The application versions are different engineering proposals. Each has a task, an embodiment, domain equations, a mapping from the supplied operators, a measurable output, an experiment and a future extension. A new version is justified by a different mechanism or contract.

```
formal source -> typed contract -> application model  
-> implementation -> observed evidence
```

Formalizing the formalization

The transformation from a source statement to an executable or measurable claim is itself recorded: source locator, interpretation, domain, equation, assumptions, proof obligation and evidence. Ambiguous source meanings remain separate until an explicit adapter relates them.

Meaning of continuing operation

In a digital system, continuation means successive valid transitions within declared memory and execution budgets. In a physical system, it means operation sustained by accounted energy and material flows. Ideal conservative models and physically powered devices have different continuation conditions; page 14 states those conditions.

Evidence status of this edition: a formal design synthesis with newly executed software and calculation checks. The application integrations are proposed architectures. The cited hardware and material studies establish bounded component evidence, and do not establish that these seven integrations have been built.

02 / NAVIGATION

Reading map

Pages	Purpose
4-6	Source authority, typed formalization and explicit source-to-contract mapping
7-10	Complete state, causal schedule, incompatible operators and geometry
11-14	SEED contract, replay, mathematical obligations and sustained operation
15	Application portfolio: what each V actually does
16-17	V01: seed-reproducible worlds and interactive symbolic spaces
18-19	V02: analog material memory and reservoir computation
20-21	V03: adaptive robotic surfaces
22-23	V04: programmable wave substrates
24-25	V05: feedback-controlled fabrication
26-27	V06: regenerative habitats and resource cycles
28-29	V07: distributed embodied systems and digital twins
30-35	Composition, feasibility, conformance, calculations and reconciliation
36-38	Source fingerprints, primary references and embedded package

Source locators use physical PDF page numbers. WANTWOMBAN has an unnumbered cover, so its printed page number is one less than its PDF page number. atomOS M1 and K1 use matching PDF and printed page numbers. UGTS PDF page 4 begins printed page 1, so its printed page numbers are three less than its PDF page positions.

V01-V07 belong to this PM 1.0 document. They do not rename atomOS v3.6, K1, UGTS-KC 3.6 or WANTWOMBAN 1.0.

Source and evidence discipline

ID	Supplied source	Contribution
S1	WANTWOMBAN 1.0 / 25 pages	64-bit controller; G0/Z0 OTAN2; numerical pinion; thermal/optical feedback
S2	atomOS v3.6 M1 / 43 pages	Typed product state; ASA/NA; JK; observations; transactional assembly; packed storage
S3	atomOS K1 validation / 8 pages	Reported CUDA word-epoch execution and bounded texture-retention experiment
S4	UGTS-KC 3.6 / 19 pages	Literal registry; geometry; ordered hinges; guarded topology; lineage; bounded linguistic profiles

Claim tags used throughout

[S] retained source definition or source-reported result. **[D]** new definition or design choice in this edition. **[L]** established law within its stated setting. **[M]** parameterized domain model. **[P]** derived consequence with assumptions. **[E]** evidence from an identified execution or experiment. **[H]** future integration hypothesis requiring investigation.

These tags describe different kinds of claims; they are not interchangeable levels of confidence. A proof of a digital mask identity cannot serve as evidence of actuator efficiency. A calibration fit cannot establish a universal mathematical identity.

Source instructions, historical prompts, commands and personal labels are documentary content. Only the current user request authorizes this synthesis. Personal attribution is metadata; it is not a physical constant, seed default, authorization token or scientific validation.

All 95 source pages were reviewed. The four original PDF byte streams are preserved as embedded witnesses. Their SHA-256 fingerprints appear on page 36. Earlier documents cited inside those sources are not counted as separately reviewed witnesses.

The formalization is an object

[D] Let a source statement be a record s with document hash, page, exact subject and context. A formalization step N produces a candidate contract K and a disposition record e . N is an editorial and mathematical construction; arbitrary natural language has no assumed unique automatic interpretation.

```
N(s, interpretation) = (K, e)
K = (Types, State, Inputs, Init, Step, Readout,
Assumptions, Invariants, Failures, Resources,
Numerics, Encoding, Provenance, Evidence)
```

Every operator carries domain and codomain, dimensions, frame/chart, totality conditions and effects. Its result is **Value(v)**, **Undefined(reason)**, or **Rejected(reason)**. An unavailable optional observation may coexist with a committed transition. A rejected transition cannot publish its candidate as authoritative.

A contract must answer six questions

Question	Required content
What exists?	All retained state, identity, units and external resources
What changes?	Transition, event order, update interval and physical evolution
What may fail?	Domain exclusions, unknown inputs, resource exhaustion and fault policy
What is preserved?	Scoped invariants and the assumptions needed for each
What can be observed?	Readout, calibration, uncertainty and comparison rule
What supports it?	Source locator, derivation, executed check or measured evidence

[D] Registry closure and type checking establish structural consistency. They do not decide all semantic propositions or prove arbitrary programs terminate. This meta-layer records unresolved obligations instead of assigning them a default PASS.

05 / EXPLICIT MAPPINGS

How the four sources connect

Source construction	PM contract role	Required boundary
S2 product state and P0-P7	Snapshot, proposals, observation, commit	M1 is a declared assembly; optional observers do not implicitly control JK/ASA
S4 definition registry and DAG	Versioned types, dependencies and explanation	Feedback crosses epochs; it does not introduce an undeclared definition cycle
S1 word-to-material loop	Physical actuator/readout adapter	Controller state, plant state and measurement are separate
S2 ASA/NA and 32-bit groups	Digital support selection	Whole-word absorption follows canonical group membership
S1 pinion and seam	Finite geometry/control transform	Rounding, clipping and seam type remain explicit
S4 hinge/support/guard	Geometry and event contracts	Kinematic mapping needs a separate physical force/process law
S3 K1 study	Bounded implementation evidence	The tested synthetic lane is a subset of the full engine architecture

A shared interface does not assert equivalence

[D] An adapter A names which source observable it carries, which information it discards, and which target law consumes it. The source record and adapter version remain visible in every trace. Equal names, bit widths or visual patterns are insufficient to establish the same operation.

Source basis: S1 PDF pp. 4, 6-12, 20-24; S2 pp. 5-7, 17-18, 23-31, 35, 38; S4 pp. 6-9, 12-14, 18-19; S3 pp. 1-8.

Complete state and physical types

```
X_n = (D, C_n, Q_n, H_n, B_n, L_n, I_n)
Z(t) = (physical fields, material history, resources)
Y_n = Readout(Z(t_n), sensor state, calibration)
U_n = Decode(C_n, Q_n, Y_n, profile)
```

[D] D is the immutable definition/configuration registry. C is digital controller state; Q is event/frontier state; H is per-stream seed/counter state; B is budgets; L is lineage and event evidence; I is the cursor into external inputs. Z is optional for a purely digital family. No single packed word is presumed to hold the complete system.

Type	Meaning	Cannot silently replace
Bit / Word<w>	Boolean / bounded integer payload	Temperature, geometric distance or continuous state
Identity / Lineage	Durable reference / ancestry	Occupancy support or coordinate equality
Angle / LogRadius	Frame-qualified radian / log of length ratio	Unscaled index difference
Temperature / Energy	Kelvin / joule	Register code or count of bit flips
Evidence / Model	Observed record / conditional prediction	Each other

[D] Quantities carry both SI dimension and semantic kind. Angles and ratios can have dimension one while requiring different adapters. Affine scales, such as Celsius to kelvin, require an offset as well as a scale. A code-to-position map must declare its origin, reference length and quantization.

S1 PDF pp. 4-8, 15; S2 pp. 8-13, 20; S4 pp. 6-9. Semantic kinds for quantities of dimension one accord with the BIPM vocabulary [R6].

One causal epoch

Stage	Effect
0 / resolve	Validate referenced definitions, units, numeric modes, inputs and available budgets.
1 / snapshot	Freeze the state and samples used by this epoch; assign input and event IDs.
2 / propose	Evaluate selected geometry, rules, seed-derived stimuli and domain predictions.
3 / guard	Check support, compatibility, invariants, domain restrictions and actuator limits.
4 / commit	Publish the accepted digital transition once; record rejected proposals separately.
5 / actuate	Send a bounded command to the physical adapter; record acknowledgment and actual command.
6 / evolve	The plant evolves during the declared interval, with delays and external disturbances.
7 / observe	Acquire timestamped readback; append lineage/evidence for the next epoch.

```
C_(n+1) = T(C_n, Y_n, input_n; D)
Z_(n+1) = Flow(Z_n, actual_U_n, disturbance_n, dt_n)
Y_(n+1) = Measure(Z_(n+1))
```

[D] All right-hand inputs belong to known snapshots or future evolution with specified initial conditions. Simultaneous events have a declared tie-break or coupled solver. A minimum positive sample interval or a proven event-accumulation policy prevents infinitely many commits in finite time.

Physical effects cannot be rolled back by restoring a word. If an actuator applied energy before a communication failure, recovery starts from the measured physical state. Digital atomic commit and physical command acknowledgment are separate events.

This schedule is a new integration of S2 snapshot/commit, S4 named phases and S1 sampled feedback. S4 pp. 12-13 use one-based prose and a zero-based diagram; PM uses the explicitly named stages above.

Keep incompatible operators named

Namespace	Definition and domain
WB.OTAN2.GO	$((r \& 2047) \text{ XOR } (O \ll 10)) \& 2047$; O is a bit. Produces an 11-bit phase code.
WB.OTAN2.ZO	$((((r \& 2047) \text{ XOR } O) \ll 10) \& 2047)$. Only outputs 0 or 1024.
ATOM.OTAN2.ratio	$\text{atan}(d\text{phi}/d\text{rho})$, $d\text{rho} \neq 0$; dimensionless increments share a specified chart/interval.
ATOM.OTAN2.directed	$\text{wrap_2pi}(\text{atan2}(d\text{phi}, d\text{rho}))$; nonzero vector; range $[-\pi, \pi]$. Explicit opt-in completion.
UGTS.logpolar.angle	$\text{atan2}(y, x)$ in a declared geometric chart, with an explicit singular-core policy.

[S] The first two are finite code transformations; the next two observe increment orientation; the last is a coordinate angle. A physical decoder may map a WB phase code to $2 \cdot \pi \cdot \text{code} / 2048$. That map does not make the code operator equal to the geometric angle of an arbitrary point.

Hadamard has two meanings here

[S] WANTWOMBAN defines the numerical transform $A_{\text{phi}} = \text{diag}(\text{phi}, 1/\text{phi}, 1, -1) H_4$, where H_4 is a normalized 4 by 4 Hadamard matrix. atomOS retains a Boolean XOR blend. These have different domains, invariants and implementations.

[D] Missing values retain reason codes. For example $d\text{rho}=0$ is undefined for the literal ratio profile even when the directed completion has a value. A fallback changes the operator profile and must be selected explicitly.

S1 PDF pp. 7-9; S2 pp. 19-22, 40; S4 pp. 5, 8. These names prevent a source ambiguity from silently becoming a physical claim.

09 / SEMANTICS

Geometry, logic and topology

```
r = r0 * 2^(Delta_rho * signed11(rho_code))
theta = 2*pi*theta_code/2048
WB.seam: (rho,theta,kappa) ->
(-rho, (1024-theta) mod 2048, kappa XOR 1)
```

[S] WB radial seam inputs use the symmetric range -1023..1023, so applying the seam twice restores the fields. The chosen reflection supports the quotient semantics; reciprocal-radius inversion is a separately attached involution. A physical seam requires an actual site/port map. Scalar temperature and material fluxes must satisfy the physical boundary conditions at connected ports.

```
a = x & M_A & valid; n = a & M_N
h = popcount(n & boundary)
y = 0 if h > 0 else n
q_next = (J & (1-q)) | ((1-K) & q)
```

[S] atomOS absorption clears a whole logical word when a selected boundary bit is hit. Its groups are 32 bits. The WB control word is 64 bits. They are separate types. JK updates use the old q and explicit J/K inputs; occupancy is not automatically a heater command.

Transfer only the invariant actually established

[S/P] The real numerical pinion preserves four-dimensional volume magnitude, not arbitrary length or physical energy. Fixed-point clipping loses that exact invariant. UGTS implicit fields provide sign/boundary semantics; exact distance requires a stronger capability. A regular m-gon is not a general (m-1)-simplex for m>3.

S1 PDF pp. 6-12, 21; S2 pp. 11, 14, 17-18, 27, 35; S4 pp. 7-9, 12-14. Quotient background: Hatcher [R5]. The online WB no-coordinate-squaring restriction applies to that kernel; domain energy laws retain their required arithmetic.

10 / PPM-SEED-v1

SEED / Exact digital construction

[D] This new reproducibility contract supplies digital initialization and experimental stimuli. H is SHA-256; || is byte concatenation; U16/U64 are unsigned big-endian fields. Every displayed \0 is a single NUL byte. Lengths count bytes; overflow is rejected.

The core seed S is exactly 32 bytes: an integer in $[0, 2^{256})$, persisted as 64 lowercase hex digits. A decimal adapter accepts canonical ASCII unsigned decimal only. Optional text input uses strict UTF-8, with no normalization, trimming or case folding, and hashes a length-framed string:

```
S_text = H("PPM/TEXT/v1\0" || U64(len(t)) || t)
m = H("PPM/MANIFEST/v1\0" || U64(len(C)) || C)
K = H("PPM/SEED/v1\0" || S || m)
```

C is a constrained canonical JSON manifest with exactly nine root fields: **schema**, **family**, **sub_v**, **numeric_mode**, **seed_hex**, **sources**, **assets**, **algorithms**, **parameters**. schema is ppm-seed-v1; seed_hex encodes S. Source/asset lists contain unique, ID-sorted {id,sha256} records. Profile IDs and sampling domains are lower ASCII slash-separated identifiers.

Canonical encoding uses UTF-8 without BOM, recursively sorted ASCII keys, no token whitespace and preserved array order. Keys match `[a-z][a-z0-9_]*`. Values may be null, booleans, Unicode scalar strings, lists or objects. Reject duplicate keys, invalid Unicode, JSON numeric tokens and nonfinite values. Numeric quantities use schema-typed strings. Strings follow RFC 8785 escaping, preserving their Unicode sequence; this is a deliberately restricted subset [R7, R8].

The application schema fixes units, constant representations, initialization, algorithm versions and resource limits. Source/asset hashes bind exact bytes; no field may contain its own manifest/output digest. Host paths and incidental timestamps belong in a receipt unless they affect the mathematics.

The embedded seed_reference_v1.py fixes all byte rules, rejects and fixture values. Its generic validator checks seed/manifest structure; a complete application must also resolve referenced contents and validate its own parameters. The custom stream is a new definition, not a standardized random-bit generator or a source of physical randomness.

11 / PPM-SEED-v1

SEED / Streams, evidence and replay

```

B(d,c) = H("PPM/BLOCK/v1\0" || K || U16(len(d))
|| ASCII(d) || U64(c))
x = unsigned_big_endian(B(d,c)[0:8])
L = 2^64 - (2^64 mod n)
advance c after every draw; accept x < L;
output x mod n; otherwise retry

```

[D] Domain d has 1..65535 ASCII bytes; each slash-separated component matches `[a-z0-9_-.]+`. Counter c is $0..2^{64}-1$. The u64 sampler consumes one 32-byte block, uses its first eight bytes and discards the rest. For n in $1..2^{64}$, record every rejected draw and advance only that domain. Use arithmetic wide enough to represent 2^{64} . A next counter equal to 2^{64} is terminal and is never hashed. A unit-grid API in a distinct domain returns $(x \gg 11) * 2^{-53}$ in $[0,1)$.

Use stable logical identities and sampling-site names, such as geometry/instance/0/position. Worker IDs and arrival order cannot define random streams. The range mapping removes modulo bias conditional on uniform input words; this does not prove a finite public seed set has perfect statistical distribution.

```

T0 = H("PPM/TRANSCRIPT/v1\0" || m
|| U64(len(X0)) || X0)
T_(i+1) = H("PPM/EVENT/v1\0" || T_i || U64(i)
|| U64(len(E_i)) || E_i)

```

X0 is exact family-defined initial-state bytes; E_i is a constrained-canonical event object. Record input/measurement content, logical order, draw counters, outcomes and state checkpoints. A receipt binds terminal count and chain head. Replay recomputes transitions and reports the first divergence; an unsigned hash alone establishes neither execution truth nor authorship.

Replay modes: exact integer/fixed-point state; explicitly pinned binary64 environment; numerical tolerance with units and exact discrete statuses; or measurement compatibility. Each is a different claim [R22]. Transcendental implementations, operation order and physical sensor novelty require their own records.

Physical timestamps, calibration, actual commands and measurements are never regenerated retroactively from a seed. WB uses ties-away rounding; a new fixed-point profile must declare its own rounding. The SEED reference does not implement WB, and adapters must preserve the selected source arithmetic.

12 / DERIVATIONS

What the formal claims require

P1 / Conditional digital replay

[P] Fix the complete initial state, semantic descriptor, seed bytes, input transcript, numeric rules and event order. If every enabled transition is a deterministic function on that data, every successive state and canonical output is identical. **Proof:** identical epoch inputs give identical proposals, guards and commit; induction gives the result. Equal seed alone does not fix the other premises.

P2 / Invariant preservation

$I(X_0)$ and $[I(X) \text{ and } \text{Guard}(X,u) \Rightarrow I(T(X,u))]$
 imply $I(X_n)$ for every committed n .

[P] Induction proves this statement for the digital state. A physical invariant additionally needs a property of Flow and the actuator/readout interface. Checking only T leaves the intervening continuous trajectory unconstrained.

P3 / Explicit cross-domain refinement

$R(x,z)$ and permitted input u
 $\Rightarrow R(T(x,u), \text{Flow}(z, \text{Decode}(u), dt))$
 with declared observable error $\leq \epsilon$.

[D/P] R relates a model state x and an implementation or physical state z . The relation may require a quantization cell, tolerance, operating domain and finite horizon. An exact digital adapter uses equality of decoded observables. An approximate physical adapter needs error propagation and measured parameter uncertainty.

P4 / Continuous event validity

[D] A geometric guard must have a certified root/crossing within the search interval, or an explicit no-event/unknown result. Define initial contact, tangency, simultaneous roots and search exhaustion. Samples that miss a boundary cannot establish continuous clearance.

New derivations extend the invariants and caveats in S1 PDF pp. 4, 12; S2 pp. 30-35; S4 pp. 12-14. They are scoped arguments, not a proof of every implementation.

13 / ENERGY AND CONTINUATION

The perpetuation condition

$$E(t_1) - E(t_0) = \int [P_{in} - P_{useful} - P_{loss}] dt$$

Repeated cycle: $\int P_{in} dt$
 $= \int (P_{useful} + P_{loss}) dt$

[L/P] For a repeated cycle whose complete stored-energy state returns to its starting value, net supplied energy equals delivered output plus losses. Any harvested light, heat gradient, fuel, pressure, gravity change or replenished material belongs in the boundary account. The sign convention follows energy conservation [R1].

A finite reserve gives a finite bound

$$P_{in} = 0, P_{useful} + P_{loss} \geq p_0 > 0, E \geq E_{min}$$

$$\Rightarrow t_{run} \leq (E_{start} - E_{min}) / p_0$$

[P] Integrating the energy inequality proves the bound. If losses are zero in an ideal model, persistent motion may exist without useful net extraction; extracting energy changes the reserve. A topology label, seed, matrix determinant or reset cannot cancel an energy term.

Digital recurrence and physical endurance

[P] An autonomous deterministic finite-state core eventually revisits a state. Periodic external input may be included by augmenting the core with its phase. An unbounded counter or growing trace is additional state; a finite implementation must stop, checkpoint, or declare wrap behavior. Eventual digital recurrence does not establish an infinite hardware lifetime.

[D] Every physical application therefore states a mission horizon H , input-resource schedule, storage bounds, degradation model and service condition. Its continuation claim is conditional on those quantities remaining inside the admitted envelope.

S1 PDF pp. 13-19 and S4 p. 16 already distinguish physical energy from symbolic operations. PM uses the title for sustained recurrence and explicitly retains the energy balance. No zero-input, positive-output perpetual device is established here.

14 / PORTFOLIO

Seven application architectures

V	Concrete application	Dominant embodiment	Future extension
01	Seed-reproducible worlds and symbolic interfaces	Digital geometry, logic and GPU evaluation	Persistent spatial environments spanning devices
02	Material memory and temporal computation	Analog thermal/electrical dynamics + digital readout	Computing skins and distributed material inference
03	Adaptive robotic surfaces	Physical hinges, compliant actuators and sensing	Deployable structures that reshape under feedback
04	Programmable wave substrates	Analog acoustic/photonic propagation + digital tuning	Distributed wave-based sensing and computation
05	Feedback-controlled fabrication	Material deposition, process sensing and guarded motion	In-situ repair and adaptive manufacturing
06	Regenerative habitats	Energy, water and thermal plant + scheduling	Resource-constrained remote/space installations
07	Distributed embodied systems	Networked physical agents + digital twins	Coordinated swarms with reproducible planning

[D] These are application contracts, not seven claims of completed products. Each combines particular source mechanisms with a declared domain model. The original source implementations remain identifiable within the combination.

SEED has a concrete role: it can generate digital geometry, stimuli, task assignments and replayable simulations. Physical initial state, disturbances, calibration and sensor observations remain measured or explicitly modeled inputs. A seeded simulator of an analog circuit remains a digital simulator.

On the following pages, [H] marks the future integration rather than an established result. Component research is cited at the point where it supports the proposed physical mechanism.

V01 / Seed-reproducible worlds

Task: regenerate and interact with a spatial environment from a compact seed plus a complete ruleset, while preserving object identity, causality and edit history across visits and devices.

[D] The runtime expands a bounded UGTS grammar into typed geometry; durable lineage addresses locate objects. Named SEED streams select branches, material IDs and procedural parameters. WANTWOMBAN pinion/phase rules provide one optional structural motif. atomOS produces and filters occupancy with explicit word-group absorption, while its observer bank records selected geometric invariants.

```
node = Expand(grammar, parent_id, branch, seed_draw)
geometry = Realize(node, chart, frame, units)
visible = Admit(Emit(geometry), ASA_profile)
world_next = Commit(world, event, accepted_edits)
```

Port	Contract
Input	Seed, grammar hash, scene descriptor, event transcript, resource budget
State	Definition DAG, object identities, frontier, editable scene state, counters, ledger
Output	Canonical scene/occupancy records plus a chosen visual or interaction adapter
Device adapter	Linear/Morton texture storage must decode to the same canonical 32-bit logical groups

The resulting environment is editable state. Replacing the grammar or applying an edit creates a versioned event and invalidates only declared dependents. A seed is a compressed generative recipe; it cannot encode arbitrary unrecorded human edits.

A linguistic spatial interface

[D] UGTS Dutch number profiles can map a spoken/place-value graph to pulse geometry while retaining the original integer and lexicon reference. For 23, a four-pulse polygon may visualize the declared segmentation. It is a user-interface mapping, not a new numerical value.

Source mechanisms: S2 pp. 8-12, 17-18, 24, 27-31; S4 pp. 6-14; S1 PDF pp. 7-12.

V01 / Reproduction and limits

Minimum demonstrator

[D] Generate a bounded scene with branch depth, node count and work budget declared in its manifest. Include ordinary objects, a Klein-routing portal, one linguistic chart and a local edit. Export the canonical object/occupancy records before rendering. Regenerate on two independently implemented paths.

Question	Falsifiable criterion
Exact reproduction	Canonical integer states and object IDs match for fixed descriptor, seed and transcript.
Storage portability	Linear and Morton decoders return identical logical words, including tail validity and zero padding.
Edit persistence	Replaying the recorded edit produces the same edited object and declared dependent changes.
Observer behavior	Changing optional diagnostics leaves admitted geometry unchanged in the noninterference profile.
Useful compression	Measure descriptor + seed + edits + cache storage against a materialized-scene baseline at equal output quality.

[D] Fixed-integer transforms support byte equality. Floating geometry uses a declared tolerance and environment; image equality needs pinned rasterization, shading and display conversion. Seed determinism is not a claim of identical GPU scheduling or rendered pixels.

Future extension and decisive gap

[H] A persistent spatial environment could regenerate across headsets, workstations and distributed simulation workers, exchanging sparse edits and measurement events. The gap is demonstrated total cost and stable semantic interoperability as rulebooks evolve.

Keep the architecture only if it earns its complexity. Compare it with conventional scene graphs, procedural generation and caching using equal visual fidelity, edit semantics, latency and memory limits. No source proves a universal speed or compression advantage.

17 / ANALOG APPLICATION

V02 / Material memory and computation

Task: classify or predict a time-dependent signal by letting actual material dynamics retain and mix recent input history. The useful output is a temporal feature vector read from the device.

[D/M] The reference architecture is a small array of electrically driven, thermally coupled resistive-memory cells. A lower-complexity physical experiment can use heater/thermochromic cells with a separate observation law. Heat transport follows its balance [R2]; colour and material phase need separate calibrated histories [R3]. These are distinct device choices. Dynamic memristor experiments support material-based temporal processing [R9]; thermal-neuristor research supports electrothermal cell dynamics and small-group interactions, while its larger reservoir networks are numerical studies [R10].

```

C_e,i * dv_i/dt = (u_i-v_i)/R_L,i - v_i/R_i(T_i,h_i)
C_th,i * dT_i/dt = v_i^2/R_i(T_i,h_i)
- G_iA*(T_i-Ta) - sum_j G_ij*(T_i-T_j)
h_i(t+) = H_i(h_i(t-), T_i; calibrated thresholds)

```

v and u are voltages [V], C_e capacitance [F], C_th heat capacity [J/K], R resistance [ohm], and G thermal conductance [W/K]. Require positive capacitances/resistances, $G_{iA} \geq 0$ and $G_{ij} = G_{ji} \geq 0$. The hysteresis state h requires a material-specific law. An ionic memristor cannot inherit this thermal law merely because both devices have memory.

The supplied architecture has concrete work to do

[D] UGTS binds each physical cell, calibration and permitted coupling to a typed graph. atomOS masks select addressed stimuli and record which requested batches were admitted. WANTWOMBAN supplies the power-to-heat-to-optical-feedback path when thermochromic observation is used. A physical decoder maps logical sites to electrodes; XOR selects bits while currents and heat combine by real-valued laws.

Cell calibration includes initial material state, drift, threshold history, read disturbance and actual voltage/current. An ASA rejection cannot remove heat or charge already delivered.

V02 / Readout, experiment and future

```
z_n = Normalize(measured currents or reflectance)
y_hat_n = W_out * [1; z_n]
W_out = argmin_W sum_n ||y_n-W*[1;z_n]||^2
+ lambda*||W||_F^2
```

[D] This is a proposed supervised readout adapter. Training targets, split, scaling, regularization and sample timing are fixed before evaluation. Temporal memory must be measured: nonlinearity and hysteresis can preserve unwanted history or yield unstable responses.

Minimum demonstrator

Characterize four cells with isolated steps, relaxation, threshold sweeps and pairwise coupling measurements. Apply sequences with the same pulse count in different temporal orders. Train a simple readout on one set and evaluate held-out order classification or delayed-XOR targets on another. SEED fixes the stimulus design and split; it does not initialize the material without a verified preparation procedure.

Comparison	Measurement
Memoryless sensor	Does the physical state add predictive information about past inputs?
Uncoupled cells	Does the specified coupling improve the task at matched input energy?
Digital recurrent baseline	Compare task error, latency and full driver/readout/computation energy.
Repeated physical preparation	Measure response dispersion, drift and cycle-to-cycle variation.

Future extension and decisive gap

[H] A computing skin could combine slow thermal memory, faster volatile devices and optical readout. The UGTS registry can describe different latency and reset contracts within one graph. The unresolved issue is whether that integration delivers useful fading memory, calibrated reproducibility and a total-system advantage over ordinary electronics.

Reservoir computation is a use of dynamic memory. It does not imply an energy reservoir that repeatedly supplies useful work without replenishment.

19 / EMBODIED APPLICATION

V03 / Adaptive robotic surfaces

Task: reshape a surface to cradle an object, redirect a small load or provide a slow tactile/shape display, while monitoring its physical state.

[D] A four-cell baseline combines liquid-crystal-elastomer bending elements, compliant heaters, thermochromic patches, thermistors and camera-visible pose markers. Published work demonstrates integrated LCE heating/electronics and LCE-thermoelectric position control [R11, R12]. Those component results support an actuator choice; they do not establish the proposed tiled surface.

```
C_i*Tdot_i = P_i - G_iA*(T_i-Ta)
- sum_j G_ij*(T_i-T_j) - P_tm,i
tau_q,i*qdot_i = q_eq,i(T_i,load_i,h_i)-q_i
tau_a,i*adot_i = a_eq,i(T_i,h_i)-a_i
```

[M] q is bending angle [rad], a is colour fraction, h is retained history, and τ values are positive fitted response times. q_{eq} depends on load and history. P_{tm} accounts for thermal-to-mechanical exchange in a coupled energy model; a reduced fit must bound any neglected mechanical term.

[D] UGTS ordered hinges map cell states to surface pose. Support limits local geometry work; compatibility excludes failed cells from new actuator commands while retaining their passive thermal/mechanical coupling; guards admit target changes under measured state and limits. The WB feedback loop becomes a shape-control loop through an explicit pose readout and controller adapter.

```
T_tile_i(q) = ordered product along rooted path to tile i
P_probe = P_bias + deltaP * sign(H4_column)
```

[D] The surface is assembled from per-tile transforms with connection/contact constraints. The four Hadamard sign columns provide orthogonal small-signal probe patterns. Choose bias and δP so all powers remain within $0..P_{max}$. Measured responses identify a local coupling model; orthogonal commands do not guarantee nonlinear identifiability.

Colour, pose, temperature and load remain separate observations. Powered cooling requires its own electrical input and rejected-heat ports.

20 / EMBODIED APPLICATION

V03 / Motion, control and endurance

When a geometric hinge becomes mechanics

$$\begin{aligned} M(q) \ddot{q} + C(q, \dot{q}) \dot{q} + g(q) + D \dot{q} \\ = B \cdot u + J_{\text{contact}}(q)^T f_{\text{contact}} \\ P_{\text{mech}} = \dot{q}^T \cdot B \cdot u \end{aligned}$$

[L/M] This full mechanics model replaces the reduced angle-response fit when detailed motion is required. Here u is generalized actuator effort, connected to temperature/history by a calibrated actuator law; it is not heater power. Inertia, damping and contact laws are required [R4]. A specified hinge gives kinematics; it does not supply those mechanical quantities. Moving pivots and frames require their full derivatives, rather than the reduced fixed-pivot formula.

Minimum demonstrator and proposed acceptance

[D] Identify isolated and coupled responses using four replaceable cells, current/voltage measurement, independent thermistors and one pose camera. Execute held-out unloaded and lightly loaded shapes. Proposed acceptance is normalized RMS pose error at most 10% of calibrated stroke and no excursion outside the tested material/load envelope.

Compare ordinary per-cell PI control with the coupled controller on identical hardware and target sequences. Before claiming added value, require a preregistered gain, for example 20% lower tracking error at no greater input energy. These are design targets, not achieved results. Assess repeatability, cooling delay and cycle life.

Future extension and alternative embodiment

[H] Larger arrays could become deployable supports, adaptive interiors or configurable optical mounts. Scaling needs measured load sharing and changing thermal contacts. If thermal actuation is too slow, a pneumatic adapter uses pressure/flow/chamber mechanics and compressor energy; it does not reuse a temperature-to-bend model unchanged.

Decisive gap: a repeatable joint map from heating, history and load to shape and colour, with a stable controller over the specified operating region. A Lyapunov or bounded-error argument must address the actual coupled dynamics.

21 / WAVE APPLICATION

V04 / Programmable wave substrates

Task: route coherent signals, mix amplitudes and perform linear filtering in a physical medium before digitization. A reference realization uses a tunable photonic mesh; a phononic network has a different transducer and state model.

[M] Let a and b be complex power-normalized amplitudes $[\sqrt{W}]$. S maps four input channels to four output channels; reflection, reverse propagation and loss ports are accounted separately. θ is tuning configuration and T device temperature. Published programmable silicon meshes and feedback configuration establish relevant component mechanisms [R13, R14].

```
b(omega) = S(omega;theta,T)*a(omega) + noise
M(kappa,psi) = [[sqrt(1-kappa), i*sqrt(kappa)],
[i*sqrt(kappa), sqrt(1-kappa)]]
* diag(1, exp(i*psi))
```

[D/M] M is one ideal two-channel transfer convention; κ lies in $[0,1]$ and ψ is phase [rad]. Real calibration adds loss and phase offsets. Hold tuning and temperature quasi-static during a measurement; time-modulated frequency conversion needs a separate model and tuner-work account. Tuning selects bounded electrical commands; coherent readout measures amplitude and phase. Power-only measurements $|b|^2$ discard phase and cannot alone certify signed matrix multiplication.

Source-specific compilation

[D] UGTS maps typed ports, permitted connections and ordered transformations into a device graph. atomOS selects admissible routes and records configuration/calibration. WANTWOMBAN H4 provides a concrete four-input/four-output orthogonal mixing target. The compiler must solve a bounded fit between measured and requested transfer matrices; the graph alone does not solve that inverse problem.

```
theta* = argmin_theta sum_omega
||S_measured(omega;theta)-S_target(omega)||_F^2
```

A physical crossing requires fabricated coupling or switching. A quotient label does not create it. Boolean XOR and optical interference remain distinct operations.

22 / CROSS-DOMAIN DERIVATION

V04 / A physical pinion transform

```

phi = (1+sqrt(5))/2
H4 = 0.5 * [[1, 1, 1, 1], [1,-1, 1,-1],
[1, 1,-1,-1], [1,-1,-1, 1]]
A_phi = diag(phi, phi^-1, 1, -1) * H4
singular_values(A_phi) = (phi, phi^-1, 1, 1)
Passive scattering: S^dagger*S <= I
S_target = A_phi / phi

```

[P] For passive power-normalized transfer, the largest singular value cannot exceed one [R15]; dagger denotes conjugate transpose. Therefore the unscaled pinion cannot be implemented as passive amplitude transfer at unit numerical scale. Dividing by phi gives singular values 1, phi⁻², phi⁻¹, phi⁻¹. The decoder records the numerical scale; unscaled physical gain needs a powered amplifier and a noise/energy account.

This is a concrete cross-domain constraint: H4 is a suitable lossless ideal target, whereas the golden-ratio-scaled operator needs attenuation or active gain. The fact that $|\det(A_phi)|=1$ does not establish optical passivity.

Minimum demonstrator

[D] On a calibrated four-input/four-output bench or mesh, program H4, probe basis inputs and phase-varied superpositions, and reconstruct the complex transfer matrix. Repeat with A_phi/phi. Predeclare the allowable matrix error, bandwidth and drift; report insertion loss, settling time and total tuning/readout energy on held-out inputs.

Acoustic embodiment

```

M_q*qddot + Gamma*qdot + K(theta)*q = B*f(t)
y = L*q

```

[M] Here q is displacement, f force, and mass, damping, stiffness and transduction are measured. A wave equation or elastic-mode model supplies the propagation [R16]. Optical amplitude conventions cannot be copied into this model without an adapter.

[H] Defect-aware rerouting and coupled wave/material reservoirs are future extensions. Their unknowns include fabrication variability, nonlinear memory, stable inverse tuning and whether conversion overhead erases a computation advantage.

23 / MATERIAL APPLICATION

V05 / Feedback-controlled fabrication

Task: build a branched scaffold by depositing material, measuring what was produced and admitting the next branch only after its local support is adequate.

[D] A bounded UGTS grammar defines target sweeps and candidate attachment sites. A calibrated dispenser or printer supplies material; scanning supplies actual geometry. Visual feedback in direct ink writing and vision-controlled jetting provide relevant precedents [R17, R18]. PM adds a proposed guarded branch-admission architecture.

```
h_j(k+1) = h_j(k) + eta_j*Q_j*dt/A_j - s_j
alpha_dot = k_cure(T,I_UV)*(1-alpha), k_cure >= 0
m_retained = m_in - m_waste - m_other_out
```

[M] h is local height [m], Q deposited volume flow [m^3/s], A footprint area [m^2], η measured yield and s shrinkage [m]. Lateral spreading requires a fitted footprint kernel. α is cure fraction; the displayed first-order cure law is an illustrative, formulation-specific model, not a universal resin equation.

Source mechanism	Physical role
UGTS sweeps / lineage	Target branches, material IDs and exact ancestry of deposited segments
Support / compatibility	Search attachment neighborhood; verify permitted material/contact conditions
Guard / transition	Check clearance, measured geometry and independently calibrated support state
WB causal readback	Replace optical darkness with a declared height/process observation adapter
atomOS records	Bind planned masks, actual deposition, observations and disposition

Thermal process models need reaction heat and transported-material enthalpy when relevant. No source topology rule supplies material adhesion, strength or curing kinetics.

V05 / Growth has physical history

planned -> deposited -> inspected -> accepted_support
|-> rework / rejected_support

[D] Record deposition immediately as an irreversible operation. Only its support/admission status waits for verification. Rejected material remains in the geometry, obstruction map and waste/rework history. A restored digital checkpoint cannot remove it.

Minimum demonstrator

Print two-layer lattices and Y junctions using one material, a calibrated flow source and local scanning. Establish geometry and cure limits with independent microscopy and witness specimens. Inject repeatable bounded flow disturbances, then compare the same path in open loop and feedback.

Proposed criterion	Why it matters
At least 25% lower median absolute height error with at most 10% added time	Tests whether correction is worth its measurement and planning cost
No support accepted when independent witness checks fail	Tests the guard against the physical property it represents
Mass-balance residual within declared uncertainty	Checks retained material, incoming flow and waste accounting
Guarded graph vs ordinary height feedback	Tests the contribution of support/branch semantics beyond conventional feedback

These are proposed finite-demonstrator targets. Zero failures in a small trial set is bounded evidence, not proof of universal reliability. Visible geometry and estimated cure do not by themselves establish load-bearing strength.

Future extension and decisive gap

[H] A repair robot could scan a defect, generate a bounded repair branch, deposit material and inspect the result. Further extensions include channels and soft actuator networks. The decisive gap is online adhesion and strength estimation, followed by residual-stress and access planning for the actual material and part.

25 / RESOURCE APPLICATION

V06 / Regenerative habitats

Task: coordinate heat, electrical storage, water and process schedules over a finite mission, conserving scarce resources while maintaining declared service levels. A small demonstrator can use a greenhouse or remote enclosure; a crewed habitat is a future qualification target.

```
E_dot = eta_in*P_generation - P_load/eta_out - P_leak
M_water_dot = flow_in + flow_recovered
- flow_use - flow_discharge - flow_leak
C_T*Tdot = P_heat + P_internal - G*(T-Ta)
```

[M] E is stored energy [J]; M_water is usable water [kg]. Recovered water enters from a separately balanced waste/process reservoir; for the whole habitat, internal recovery cancels. Efficiencies lie in (0,1] and every boundary flow is explicit. Recovery reduces consumable demand while still requiring power and management of waste and contamination. NASA life-support systems provide component examples of water recovery and oxygen generation [R20].

An explicit crop-water submodel

```
D_r,i = D_r,i-1 - (P-RO)_i - I_i - CR_i
+ ET_c,i + DP_i
```

[M] The FAO daily root-zone depletion balance uses water-depth units [mm]: precipitation P, runoff RO, irrigation I, capillary rise CR, evapotranspiration ET and deep percolation DP [R19]. Convert depth to volume by area and mass by density. Soil/crop parameters and boundary fluxes must be fitted or measured; enforce capacity through physical overflow/runoff terms rather than silently clipping away water.

[D] UGTS represents the process/coupling graph. atomOS logic admits scheduled tasks under resource masks and explicit whole-group semantics. WANTWOMBAN supplies a local thermal loop. Source cycle labels become timestamped process inputs; no biological outcome follows from their names.

The retained source gestation/menstrual labels remain symbolic or recorded channels. This application introduces no fertility or medical prediction.

V06 / A finite mission contract

```
min_schedule sum_k [w_E*energy_k + w_W*water_loss_k
+ w_S*service_deviation_k]
subject to storage, flow, temperature and task constraints
```

[D] This is a proposed finite-horizon scheduling objective, with named weights and a chosen solver. Forecasts, actuator limits, disturbance bounds and fallback priorities are part of the contract. SEED can generate reproducible weather/load scenarios for design comparison; the deployed system uses measured resource state and actual forecasts.

Minimum demonstrator

Use a non-crewed enclosure with measured electrical storage, a thermal load and a recirculating water/process loop. Replay the same recorded weather/load episode in a simulator, then execute a bounded physical schedule. Test a predefined outage and a recovery scenario.

Check	Acceptance evidence
Resource conservation	Energy and water residuals consistent with sensor/model uncertainty
Service	Temperature and process requirements met over the declared mission horizon
Fault behavior	Priorities and degraded operating modes activated from actual measured state
Value of integrated scheduling	Compare fixed timers and a conventional scheduler on equal hardware, mission and constraints

Future extension and decisive gap

[H] Remote settlements or space installations could combine these process contracts with V05 repair and V07 distributed maintenance. The gap is validated reliability and environmental chemistry over long durations, including degradation, contaminants, resupply and repair resources.

Regenerative is a flow property, not a claim of total closure. State the fraction recovered, energy spent and replacement material required. A small recurring digital schedule is not evidence that a habitat can run indefinitely.

27 / NETWORKED APPLICATION

V07 / Distributed embodied systems

Task: let multiple mobile or fixed physical agents share a spatial model, allocate work and coordinate actions while retaining local control when communication is delayed or unavailable.

[D] Each agent stores a typed local twin: pose, uncertainty, resource state, acknowledged commands and a causally ordered observation log. A seed generates reproducible planning scenarios and tie-break choices from stable agent/task IDs. Actual location, contacts and message arrival history remain external inputs.

```
agent_i = (model_i, measured_state_i, input_log_i,
command_ids_i, pending_acks_i, resource_i)
message = (sender, sequence, model_version, timestamp,
measurement_or_proposal, uncertainty)
```

A bounded consensus component

```
xdot_i = -k*sum_j a_ij*(x_i-x_j), k > 0
V = 0.5*sum_i (x_i-x_bar)^2
Vdot = -k*sum_(i,j in edges) a_ij*(x_i-x_j)^2 <= 0
```

[P/M] For a fixed connected undirected graph with symmetric nonnegative weights and ideal scalar integrator agents with instantaneous delay-free communication, this component converges to agreement while preserving the mean. Count each undirected edge once in $Vdot$. The graph, delays and dynamics matter [R21]. Consensus on a scalar is not collision avoidance, actuator feasibility or agreement under arbitrary network partitions.

[D] UGTS supplies support/compatibility for local neighborhoods and explicitly guarded topology changes. atomOS maintains per-agent logic and provenance. WB-style feedback connects digital proposals to measured effects. A distributed observation update is not allowed to masquerade as an already executed physical command.

Command IDs and deduplication can prevent a known command from being applied twice by a particular actuator service. Reboot, lost acknowledgment and partial action still require physical-state reconciliation.

V07 / Replay meets the real world

Stale information has a physical cost

```
position_uncertainty(age) <= e0 + v_max*age
+ 0.5*a_max*age^2
```

[M] This conservative envelope applies when e_0 bounds the last position error and the subsequent speed/acceleration terms are valid for the chosen prediction convention. The guard expands obstacles/neighborhoods by the applicable envelope or stops accepting motion plans when its assumptions fail. Sensor latency and unmodeled motion must be included.

Minimum demonstrator

[D] Begin with three tabletop agents or actuated fixtures and one independent tracking system. Use fixed recorded task inputs. Inject bounded message delay, duplicates, dropout and an interrupted command acknowledgment. Keep local actuation inside a characterized envelope.

Evidence	What it demonstrates
Plan replay	Fixed manifest/seed/transcript reproduces digital proposals and tie-breaks
Physical trace	Independent measurements agree with stated tracking/error limits
Network fault trials	Command deduplication, stale-data handling and recovery work on tested failures
Ordinary controller baseline	Graph/lineage machinery improves a declared task metric rather than just adding logs

Future extension and decisive gap

[H] Inspection/repair swarms could maintain a V06 installation or fabricate with V05. A shared procedural map can reduce exchanged geometry, while observations update only affected regions. The unresolved issues are distributed model consistency, changing contacts, communication limits, resource allocation and robust local autonomy.

Two kinds of replay remain separate. Digital decisions may replay exactly from recorded inputs. Recreating the same physical trajectory requires matching initial conditions and disturbances within measured tolerances.

29 / APPLICATION COMPOSITION

A concrete composed machine

[D] Consider an adaptive inspection table: V01 generates an editable target surface; V03 reshapes four physical tiles; V02 classifies recent contact/thermal signals; V07 sends inspection tasks and reconciles observations. This is a proposed integrated demonstrator, with externally supplied electrical power.

Connection	Typed adapter	Obligation
World -> surface	Target pose [rad] and object/load envelope	Reachable pose and calibrated load constraints
Surface -> material memory	Normalized timestamped sensor sequence	Scaling, latency, missing-input policy and permitted read disturbance
Material output -> controller	Classifier score + calibration/version	Score affects control only through an explicit bounded policy
Remote task -> local agent	Versioned task/command ID	Freshness, deduplication and acknowledgment
All components -> evidence	State/input/command records	Consistent units, chronology and immutable reference hashes

Coupling order is part of the design

Freeze the latest accepted sensor sample. Evaluate the classifier and target proposal. Admit the command through physical limits. Record the digital commit, send the command and capture its acknowledgment. Acquire the next physical sample after the declared evolution interval. Sensor/compute/communication delays are part of that loop.

```
latency_loop = latency_sensor + latency_compute
+ latency_transport + latency_actuator
```

[D] If this latency exceeds the validated controller delay bound, use the declared degraded mode. A common formal language makes the dependency visible; it does not prove the composed loop stable. A coupled energy/uncertainty model and finite experiment remain necessary.

30 / APPLICATION VALUE

Feasibility and research decisions

Application	Established component basis	Decisive new integration question
V01	Procedural generation, packed masks, seeded computation	Does the combined representation reduce total storage/work at equal semantics?
V02	Material memory; measured small devices	Is useful temporal computation retained after readout/driver overhead and drift?
V03	Thermal soft actuators; position feedback	Can calibrated coupling improve loaded shape control and cycle life?
V04	Programmable photonic meshes	Can the source target be realized with acceptable loss, drift and tuning cost?
V05	Visual deposition feedback	Can online support/bond guards justify autonomous branch extension?
V06	Thermal/storage/process control and recycling	Does integration improve finite-mission resources and reliability?
V07	Networked control and consensus	Do reproducible models improve coordination despite stale physical information?

Evidence progression

[D] Separate a valid specification, a simulated design, a characterized component, an integrated bench demonstrator and a deployed application. Promotion requires new evidence for that specific level. Existing component papers do not promote a PM application directly to an integrated demonstration.

At each stage compare a simpler conventional baseline with the same useful output, uncertainty, hardware and resource envelope. Report negative results. A source-specific representation is useful when it improves a measured objective or clarifies a required contract; no benefit follows merely from combining names.

Future extensions in this edition are research directions. No novelty search, patent assessment, production qualification or application-level hardware benchmark was performed.

What was checked, and by whom

Evidence	Scope and outcome
S1 source-reported	10,000 word roundtrips; 10,000 field-isolation cases; 2,048 seam cases; 2,000 transition cases; OTAN2 profiles and selected thermal checks.
S2 source-reported	M36 assembly: 17 groups / 20,873 assertions. This is a reduced abstract audit, not a complete assembled production runtime.
S3 source-reported	212 bulk conformance runs; 26 independently checked exports; 24 final cold profiles at dictionary miss floor; finite workload/device evidence.
S4 source-reported	36 reference tests; a bounded 0..99 linguistic profile; no independent physical performance validation.
PM newly executed	46 Python SEED checks and 5 independent .NET framing/hash comparisons on the same Windows host.
PM newly executed	15 source-operator identity and cross-domain calculation checks; result details embedded in formal_checks_results.json.

[E] The new SEED checks include format/Unicode rejects, canonicalization, manifest binding, domain/counter changes, rejection-sampling boundaries and transcript mutation. The .NET implementation independently reconstructs fixed frames and hashes. This is useful same-host implementation evidence, not qualification across operating systems.

[S] S3 reports a 4 MiB dictionary distributed over 46 SM caches. At 32 bytes per sector its compulsory miss floor is 131,072. Reaching that aggregate floor during a cold warm/work/reread epoch supports bounded retention under its recorded conditions. It does not establish pinning across launches or arbitrary workloads.

No source CUDA commands or embedded historical instructions were executed for this synthesis. New checks concern the added formalization support; V01-V07 remain application proposals.

32 / CALCULATIONS

Worked, reproducible consequences

Thermal recurrence inherited from WANTWOMBAN

```
C=0.020 J/K; G=0.005 W/K; P=0.100 W; Ta=293.15 K
tau=C/G=4 s; T_infinity=Ta+P/G=313.15 K
T(t+dt)=Ta+P*u/G+(T(t)-Ta-P*u/G)*exp(-dt/tau)
```

[P/E] For the illustrative 304.15 K heating threshold, first passage from ambient is 3.19403 s. Supplied energy is 0.319403 J; sensible storage is 0.220000 J; modeled loss is 0.099403 J. The independently recalculated values satisfy the balance. These are chosen model inputs, not measured paint properties.

Passive wave realization

[P/E] $\phi = 1.61803398875$. The unscaled pinion has maximum amplitude gain ϕ and maximum power gain $\phi^2 = 2.61803398875$. Scaling by $1/\phi$ bounds every singular value by one. A computed basis-vector witness and matrix checks are in the calculation receipt.

Exact SEED fixture

[E] The standalone fixture uses seed zero, family digital, sub_v reference-1.0.0, numeric_mode u64-mod, empty source/asset lists and domain geometry/instance/0/position. Its canonical manifest and all bytes are embedded with the reference; it is not a full application manifest.

```
B(domain,0) =
1bf021f9bb711c5514a3075a26a407c7
5959e1730dbd5df9e22f294280b7622
First ten range-10 outputs: 7,6,7,4,8,4,2,7,9,8
```

The two block-hex lines concatenate to one 64-digit value. These outputs depend on the entire fixture manifest and byte framing, not just the seed. Source thermal example: S1 PDF pp. 16-17.

An application has a complete contract

```
PM application = purpose + embodiment + source_adapters
+ state_schema + domain_laws + calibration
+ operating_envelope + scheduler + observables
+ resources + numeric/seed_profile + evidence_plan
```

[D] A concrete implementation binds these fields to immutable content and gives all optional features explicit enabled/disabled semantics. Unknown calibration cannot silently receive a plausible number. Simulation values must be labeled as inputs; measured values carry units, method, time and uncertainty [R23].

Record	Minimum information
Definition	ID, version, content hash, domain/codomain, assumptions, source locator and capabilities
Plant	Device/material IDs, initial/boundary conditions, calibration, response delays and resource ports
Run	Initial state, actual inputs, seed profile if used, execution order, code/math versions and limits
Event	Proposal, guard result, actual command/acknowledgment, readback and lineage
Evidence	Method, configuration, raw records, comparison rule, finite coverage and unresolved claims

When a V changes

A new application V is justified by a different task or physical/computational architecture. Within a V, incompatible changes to state, law, actuator/readout meaning or observable semantics require a new major contract version. A calibration revision or new fitted parameter set has its own identity; an editorial correction does not imply new physical behavior.

The minimal SEED module is not a full validator for these application contracts. The embedded editable source supports subsequent implementation; it does not pretend to be a driver for unspecified hardware.

Reconciliation and retained distinctions

Issue	Formal treatment in PM
OTAN2 collisions	Keep five explicitly named roles; never silently substitute atan2 for literal ratio or code mixing.
Hadamard collisions	Numerical H4 and Boolean XOR remain different; wave realization obeys passivity.
Word size / identity	64-bit WB controller, 32-bit ATOM sink groups, identity digest and lineage are separate types.
Infinite nesting	Unbounded symbolic addresses require storage for every finite execution prefix; finite runs enforce budgets.
Topology as physics	A seam defines routing/identification; hardware coupling and constitutive laws remain explicit.
Source pipeline numbering	Use named stages and explicit index bases, preserving actual causal order.
Pulse simplex wording	Use point/segment/regular polygon for the supplied planar embedding; higher simplex needs another map.
Hinge derivatives	Fixed-pivot formula requires fixed pivot/source point; moving components require additional derivatives.
Distance and pruning	Sign fields are not automatically exact distances; culling needs a descendant-containing bound.
Physical rollback	Deposited heat, charge and material remain in measured history after a digital rejection.
Biological/numerical labels	Use as declared symbols or measured channels; no mechanism follows from lexical or count coincidences.

This table records interpretation and engineering additions instead of silently rewriting the witnesses. Relevant loci: S1 PDF pp. 7-12, 20-21; S2 pp. 10-14, 17-22, 27, 31-35, 40; S4 pp. 5-14, 16-19.

35 / PROVENANCE

Exact source witnesses

S1 / WANTWOMBAN

WANTWOMBAN.pdf
25 pages / 660,794 bytes

```
cc0d81471fe33497873faa6339aa4875  
32c8861902761994d3c88b16c17ea662
```

S2 / ATOM_TOTAL

atomOS_v3.6_Total_Integrated_Engine_Tom_Klootwijk.pdf
43 pages / 361,306 bytes

```
b51651c007c560775ff1950e278789cd  
7674e131ec71a09ebd9254cebe9c19ce
```

S3 / ATOM_VALIDATION

atomOS_v3_6_Kernel_Validation_Texture_Cache.pdf
8 pages / 23,099 bytes

```
b7ed304f9c4b6c89405ba6ba0ab3bef5  
e29fc78e5f11ad3c134333320bf31a48
```

S4 / UGTS

UGTS_KC_3_6_Tom_Klootwijk.pdf
19 pages / 1,769,281 bytes

```
e21433cd8378368dafe4ae278d3c9c36  
4725e45ae7fd75b743328bd2d6003a46
```

Concatenate each pair of hash lines. Hashes identify the supplied byte streams, not independent authorship or scientific truth. The original PDFs are attached to this PDF without modification. Source-local links and historical commands remain part of their documentary content.

Primary references / 1

Consulted 15 September 2026. These sources support the stated component laws and examples. The seven PM architectures, mapping contracts and proposed experiments are new synthesis.

[R1] NASA Glenn. First Law - Internal Energy. Energy-balance convention. [Primary source](#).

[R2] COMSOL 6.3. The Heat Balance Equation. Local/integral thermal balance. [Primary source](#).

[R3] Basnec et al. (2018). Colour and phase changes of a leuco-dye thermochromic composite. Scientific Reports 8, 5511. [Primary source](#).

[R4] Tedrake. Underactuated Robotics: Multi-Body Dynamics and Lyapunov Analysis. Physical equations and scoped stability conditions. [Primary source](#).

[R5] Hatcher. Notes on Introductory Point-Set Topology, chapter 4. Quotient construction. [Primary source](#).

[R6] BIPM / JCGM vocabulary, 1.8. Quantities of dimension one retain semantic meaning. [Primary source](#).

[R7] NIST FIPS 180-4. Secure Hash Standard. Pinned SHA-256 primitive. [Primary source](#).

[R8] RFC 8785. JSON Canonicalization Scheme. PM uses a constrained subset. [Primary source](#).

[R9] Du et al. (2017). Reservoir computing using dynamic memristors for temporal information processing. Nature Communications 8, 2204. [Primary source](#).

[R10] Zhang et al. (2024). Collective dynamics and long-range order in thermal neuristor networks. Nature Communications 15, 6986. [Primary source](#).

[R11] Multimaterial Printing of Liquid Crystal Elastomers with Integrated Stretchable Electronics (2023). ACS Applied Materials & Interfaces. [Primary source](#).

[R12] Zadan et al. (2022). LCE with integrated soft thermoelectrics for shape-memory actuation and energy harvesting. Author-institution record. [Primary source](#).

R4 stability chapter: [Lyapunov Analysis](#). R7 reference-check vector: [NIST SHA-256 abc example](#).

37 / RESEARCH AND PACKAGE

Primary references / 2

[R13] Perez et al. (2017). Multipurpose silicon photonics signal processor core. Nature Communications 8, 636. [Primary source](#).

[R14] Perez-Lopez et al. (2020). Multipurpose self-configuration of programmable photonic circuits. Nature Communications 11, 6359. [Primary source](#).

[R15] Ullrick et al. (2023). Wideband parametric baseband macromodeling of linear and passive photonic circuits. Scientific Reports 13, 15407. [Primary source](#).

[R16] MIT OpenCourseWare, 18.303. The 1-D Wave Equation. Propagation and boundary models. [Primary source](#).

[R17] Piovarci et al. (2022). Closed-Loop Control of Direct Ink Writing via Reinforcement Learning. Author project/paper. [Primary source](#).

[R18] Buchner et al. (2023). Vision-controlled jetting for composite systems and robots. Nature 623, 522-530. [Primary source](#).

[R19] FAO Irrigation and Drainage Paper 56, chapter 8. Root-zone water balance, equation 85. [Primary source](#).

[R20] NASA. Environmental Control and Life Support Systems. Water recovery and oxygen generation. [Primary source](#).

[R21] Olfati-Saber and Murray (2004). Consensus problems in networks of agents with switching topology and time-delays. IEEE TAC 49(9). [Primary source](#).

[R22] NVIDIA. Floating Point and IEEE 754; Nsight Compute Profiling Guide. Numeric and measurement scope. [Primary source](#).

[R23] BIPM. JCGM 100:2008, Guide to the expression of uncertainty in measurement. [Primary source](#).

Embedded, editable package

Original S1-S4 PDFs; source/attachment manifests; editable Markdown and content JSON; PDF builder; full SEED contract/reference; Python and independent .NET receipts; formal calculation script and results. Hashes are unsigned integrity records. The PDF does not require any attachment to execute on opening.

The specification states equations and experiment designs. Seven complete application implementations and physical demonstrators are not included or claimed.